# Unit 13: Worked Examples

1. The points $(-9,2)$ and $(1,2)$ lie on the end points of the diameter of a circle. Write the equation of the circle.

$$\frac{-9+1}{2} = \frac{-8}{2} = -4$$

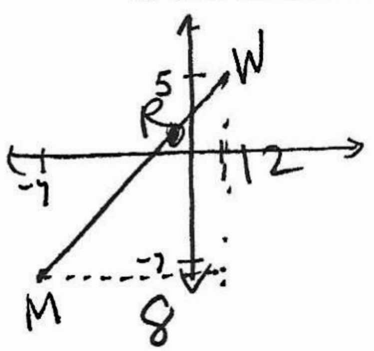
$$\frac{2+2}{2} = 2$$

$(-4,2)$
Center

$$\sqrt{(-4-1)^2+(2-2)^2}$$
$$= \sqrt{25} = 5$$
radius

$$\boxed{(x+4)^2+(y-2)^2=25}$$

2. Directed line segment $MW$ has endpoints at $(-7,-7)$ and $(1,5)$. Point $R$ is located $\frac{3}{4}$ of the distance from $M$ to $W$. Find the coordinates of point $R$.



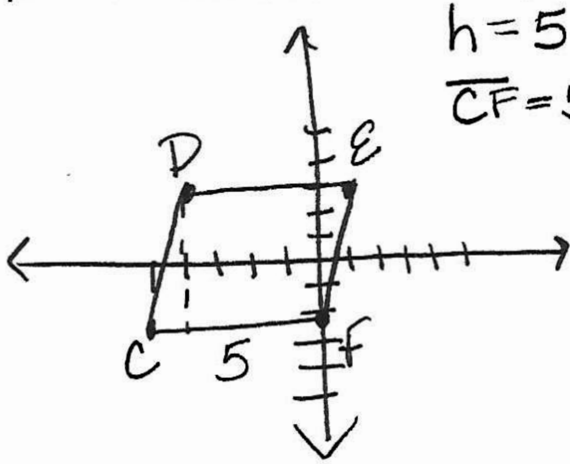

$$\frac{3}{4} \cdot 8 = 6$$

$$\frac{3}{4} \cdot 12 = 9$$

$(-7+6, -7+9)$

$(-1, 2) \leftarrow R$

3. Parallelogram $CDEF$ has vertices at $C(-5,-2), D(-4,3), E(1,3)$, and $F(0,-2)$. Find the perimeter and area of Parallelogram $CDEF$.




$h = 5$

$\overline{CF} = 5$

$$\overline{CD} = \sqrt{(-4+5)^2+(3+2)^2} = \sqrt{1+25}$$
$$= \sqrt{26}$$

$$A = \boxed{5\sqrt{26}}$$

$$P = 5 \cdot 2 + 2\sqrt{26}$$

$$\boxed{P = 10 + 2\sqrt{26}}$$